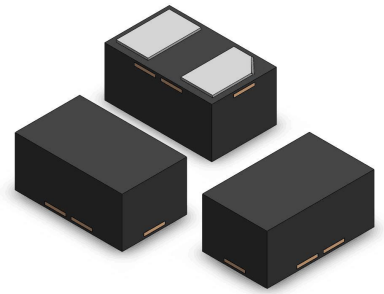
Transient Voltage Suppressors for ESD Protection

ESDXXV88D-CDN Series

Description

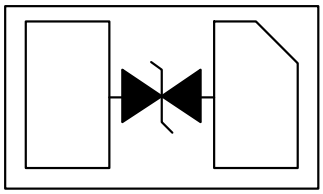
The ESDXXV88D-CDN series is designed to protect sensitive components which are connected to high-speed data and transmission lines from over-voltage caused by ESD (electrostatic discharge), CDE (Cable Discharge Events), and EFT (electrical fast transients).



Feature

- 210 ~ 285Watts Peak Pulse Power per Line (tp=8/20μs)
- Protects one Bidirectional I/O line
- Low clamping voltage
- Working voltages: 3.3V ~ 36V
- Low leakage current
- IEC61000-4-2 (ESD)±30kV (air), ±30kV (contact):3.3V-15V
- IEC61000-4-2 (ESD)±25kV (air), ±15kV (contact):18V-36V
- IEC61000-4-4 (EFT) 40A (5/50ηs)

Functional Diagram



Applications

- Cell Phone Handsets and Accessories
- Notebooks & Handhelds
- Notebooks, Desktops, and Servers
- Portable Instrumentation
- Peripherals
- Pagers

Mechanical Data

- SOD-882/DFN1006 (1.0x0.6x0.5mm) Package
- Molding Compound Flammability Rating : UL 94V-O
- Weight 0.5 Milligrams (Approximate)
- Lead Finish : Lead Free

Mechanical Characteristics

Symbol	Parameter	Value	Units
P_{PP}	**Peak Pulse Power (tp=8/20μs waveform)**	210 ~ 285	Watts
T_L	**Lead Soldering Temperature**	260 (10 sec.)	ºC
T_{STG}	**Storage Temperature Range**	-55 to +150	ºC
T_J	**Operating Junction Temperature Range**	-40 to +125	ºC

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Electrical Characteristics (@ 25℃ Unless Otherwise Specified)

Part Number	Device Marking Code	Stand-Off Voltage V_{RMW} (V)	Breakdown Voltage V_{BR} (Min.) (V)	Test Current I_T (mA)	V_C @1A (Max.)	V_C		Maximum Reverse Leakage I_R @V_{RWM} (uA)	Typical Junction Capacitance (pF)
						(Max.)	(@A)		
ESD3.3V88D-CDN	D3	3.3	3.5	1.0	6.0	14.0	15.0	1.0	20
ESD05V88D-CDN	D5	5.0	6.0	1.0	9.8	18.0	15.0	1.0	60
ESD08V88D-CDN	D8	8.0	8.5	1.0	12.0	22.0	10.0	1.0	45
ESD12V88D-CDN	DD	12.0	13.3	1.0	17.0	28.0	8.0	1.0	27
ESD15V88D-CDN	DE	15.0	16.7	1.0	22.0	35.0	6.5	1.0	25
ESD18V88D-CDN	DF	18.0	19.5	1.0	25.0	40.0	6.0	1.0	45
ESD24V88D-CDN	DH	24.0	26.7	1.0	36.0	55.0	4.5	1.0	24
ESD36V88D-CDN	DN	36.0	40.0	1.0	50.0	70.0	3.0	1.0	20

Characteristic Curves

Fig1. 8/20μs Pulse Waveform

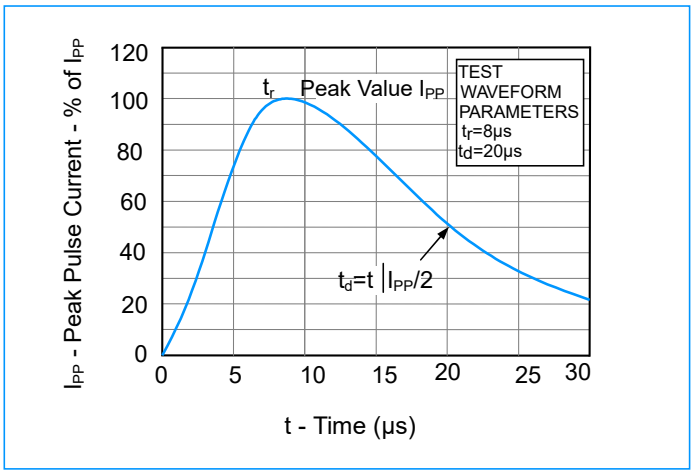


Fig2. Power Rating Derating Curve

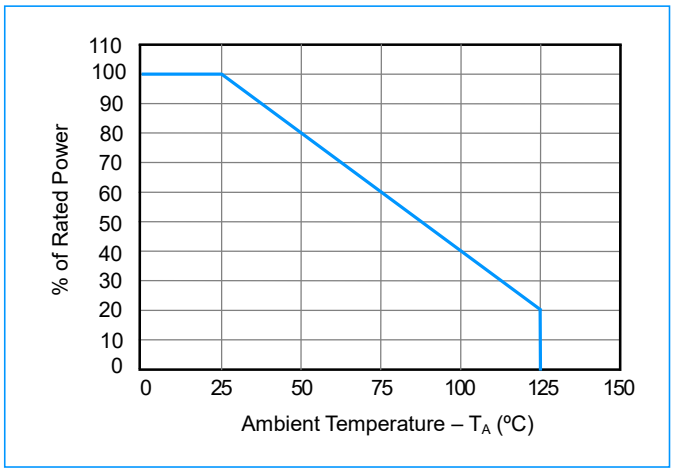
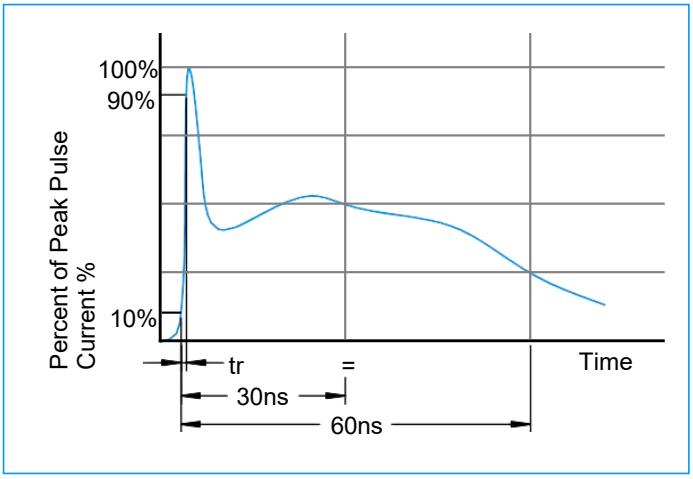


Fig3. ESD Pulse Waveform(according to IEC61000-4-2)



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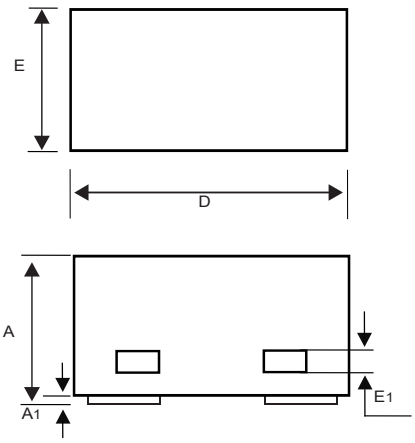
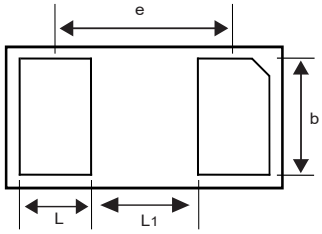
SOD-882/DFN1006 Package Outline & Dimensions

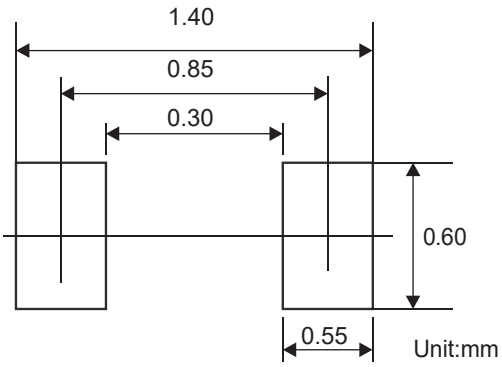
SOD-882/DFN1006





Symbol	Millimeters		
	Min	Nom	Max
A	0.450	0.500	0.550
A1	0	0.020	0.050
E1	0.013	0.063	0.113
D	0.900	1.000	1.100
E	0.500	0.600	0.700
e	0.65BSC		
L	0.150	0.250	0.350
b	0.400	0.500	0.600
L1	0.300	0.400	0.500

Suggested PAD Layout



Ordering Information

Device	Package	Quantity	Reel Size
ESDXXV88D-CDN Series	SOD-882/DFN1006	12,000pcs/Reel	7 inch

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